

Human-AI Team Accuracy via Subset Selection and Cost-Effective Collaboration

A Technical Summary of ComHAI and PLACO

1. Core Problem and Motivation

In classification tasks, neither a human nor an AI model is perfectly accurate in isolation. The central idea is to **combine** the outputs of one or more humans with the probabilistic output of an AI model to yield a joint prediction that outperforms either alone. Key observations motivating the framework:

- Humans and AI models make *different kinds of errors*, so their combination is complementary.
- A single human’s label limits the combined accuracy; involving *multiple* humans with diverse expertise helps.
- Adding more humans is **not always better** — a poor annotator can actively hurt accuracy.
- Querying human annotators has **cost**; in practice, labels from every human on every instance are not freely available.

2. Method: ComHAI — Combining Multiple Humans with AI

2.1. Human Track Records via Confusion Matrices

Every human annotator’s reliability is captured by a **confusion matrix** $\phi^{[i]}$, where $\phi_{st}^{[i]} = P(\text{human } i \text{ predicts } s \mid \text{true label is } t)$. This encodes not just overall accuracy but *which classes each human tends to confuse with which*. Diagonal entries represent correct predictions; off-diagonal entries capture systematic biases. A Bayesian estimation procedure is used to build these matrices efficiently from a modest number of labeled examples.

2.2. Correcting AI Overconfidence

Neural network classifiers are known to output probabilities that are *systematically overconfident* — the model’s stated certainty exceeds its true accuracy. Before any combination is performed, the AI model’s probability vector $m(x)$ is **calibrated** (rescaled) to produce $m^\theta(x)$, a more faithful estimate of the model’s true confidence on instance x . Only calibrated probabilities are used downstream.

2.3. Combining Human Labels with AI Output (ComHAI)

Let $m^\theta(x)$ be the calibrated AI probability over K classes, and let humans $i = 1, \dots, n$ each predict a hard label for instance x . Assuming human predictions are *independent of each other* given the true label, Bayes’ rule yields the **joint posterior**:

$$P(y=j \mid h(x), m(x)) \propto m_j^\theta(x) \prod_i \phi_{h_i(x), j}^{[i]}$$

The final prediction is the class j that maximises this expression. Intuitively: the AI’s confidence in each class is *multiplied* by each human’s historical tendency to give their observed answer when that class was the truth. A human who almost never calls “cat” on a “dog” image strongly

down-weights the “dog” prediction when they say “cat”.

2.4. Simulating Human Labels for Experiments

To test the framework, human predictions are *simulated* from the CIFAR-10H dataset. A human with accuracy Ψ on instance x predicts:

$$h^{[\Psi]}(x) = \begin{cases} y(x) & \text{with prob. } \Psi \\ t \sim p(k|x) & \text{with prob. } 1-\Psi \end{cases}$$

where $p(k|x)$ distributes errors proportionally to how often other real annotators chose each wrong class, reflecting realistic confusion patterns rather than uniform noise.

2.5. The “Too Many Cooks” Problem (Lemma 4.2)

A critical theoretical result shows that adding more humans to the combination is **non-monotone**: even if every human is individually more than 50% accurate, including all of them can *reduce* overall accuracy. The root cause: a human who guesses the wrong class with moderate probability multiplies the posterior of that wrong class, pulling the final answer in the wrong direction. This makes *subset selection* essential, not optional.

2.6. Lower Bound and the Subset Selection Objective

To guide subset selection without knowing the true label, a **lower bound on accuracy** is derived. The lower bound improves when the product

$$\prod_{i \in S} \frac{\phi_{h_i(x), y(x)}^{[i]}}{1 - \phi_{h_i(x), y(x)}^{[i]}}$$

is maximised — a product of *odds ratios*, one per selected human. Each factor exceeds 1 exactly when human i ’s answer is more likely to be correct than incorrect (given their track record). Including a human with a factor < 1 *hurts* the bound.

2.7. GreedySubsetSelection: Pseudo-Optimal Subset

Since the true label $y(x)$ is unknown at decision time, a **pseudo-label** is used:

$$y^* = \arg \max_j \prod_{i: \text{ratio}_j^{[i]} > 1} \frac{\phi_{h_i(x), j}^{[i]}}{1 - \phi_{h_i(x), j}^{[i]}}$$

The algorithm treats y^* as a surrogate for the true label and then selects all and only those humans for whom the odds ratio (computed against y^*) exceeds 1. This is computed in $O(nK)$ time — a single pass over humans and classes — with no exhaustive subset enumeration needed.

3. PLACO: Adding Cost-Awareness

3.1. Why Cost Matters

ComHAI’s GreedySubsetSelection requires knowing every human’s label on every instance before deciding whom to include. In practice, labels cost money and time. **PLACO**

(Probabilistic Labeller-Assisted Cost Optimisation) removes this requirement by *estimating* human labels before eliciting any of them.

3.2. Human Label Estimation via AI Confidence

Instead of asking a human what they would predict, PLACO *infers* the most likely label that human would give, using only:

- The AI model’s probability vector $m(x)$ on the current instance.
- The human’s confusion matrix $\phi^{[i]}$ (their historical track record).

Formally, the estimated label is:

$$h_i(x) = \arg \max_l \sum_y \phi_{l,y}^{[i]} \cdot m_y^\theta(x)$$

This is a *posterior* estimate: “given what the AI thinks is likely, what would this human most probably say?” Experiments show this substantially outperforms naive alternatives (e.g., always predicting the human’s most common answer, or random guessing).

3.3. Value Function for Human Usefulness

With estimated labels in hand, PLACO defines a **value function** $V_i(x, j)$ for each human i and candidate label j — a lower bound on the odds-ratio contribution that human i would make if included. The value accounts for:

- Whether the human is likely to be *ideal* (i.e., likely to predict the true class), proxied by their overall accuracy.
- The estimated label $h_i(x)$ and the human’s diagonal confusion matrix entries.

A human with high value is one who, based on their track record and the AI’s reading of the instance, is likely to *push the combined prediction toward the correct class*.

3.4. PLACO Greedy Subset Selection

Mirroring GreedySubsetSelection but using V_i instead of true odds ratios: the pseudo-label is $y^* = \arg \max_j \sum_i V_i(x, j)$, and the selected subset includes all humans with $V_i(x, y^*) > 1$. At least one human is always included. Only the selected humans’ *true* labels are then elicited and used in ComHAI’s combination formula — significantly reducing cost.

3.5. PLACO LP: Budget-Constrained Selection

When a hard budget B is imposed, PLACO LP formulates subset selection as an **Integer Linear Programme**:

$$\max_{e_i \in \{0,1\}} \sum_i \log V_i(x, y^*) \cdot e_i \quad \text{s.t.} \quad \sum_i c_i e_i \leq B$$

where c_i is the cost of querying human i . This allows tuning the accuracy-cost tradeoff explicitly: a tighter budget forces selection of fewer (but still valuable) humans; a looser budget recovers near-full accuracy.

4. Three-Model Hierarchy

1. **ComHAI + GreedySubsetSelection**: Optimal subset selection assuming all human labels are freely available. Sets the accuracy ceiling.
2. **PLACO Greedy**: Uses estimated labels for subset selection; elicits true labels only from the chosen subset. Matches ComHAI accuracy at a fraction of the cost.

3. **PLACO LP**: Adds a budget cap via ILP. Trades some accuracy for guaranteed cost control.

5. Experimental Results

5.1. Datasets and Models

Experiments use **CIFAR-10H** (10-class, 10,000 images with real human annotations) and **ImageNet-16H** (16-class). Multiple AI backbones are tested:

- **Custom CNN** (3 conv layers, 56.74% accuracy on CIFAR-10H) — intentionally weak.
- **ResNet-110** (93.89% on CIFAR-10H) — strong baseline.
- **CNN for ImageNet-16H** (43.40%) — weaker than CIFAR-10H CNN.

Human configurations range from 4–15 annotators with accuracies spanning 0.5–0.8, all results averaged over 10 runs.

5.2. ComHAI Accuracy Gains

- With CNN (56.74%) + 7 humans (acc. 0.5–0.8): ComHAI reaches **96.65%**.
- With ResNet-110 (93.89%) + same setup: ComHAI reaches **99.15%**.
- Naïve baselines (best human only, majority vote): plateau around 80–85%.
- “All Humans Selected” degrades severely with 15 humans, confirming non-monotonicity.
- Pseudo LB (GreedySubsetSelection) consistently achieves the best accuracy among practical methods.

5.3. PLACO Cost-Accuracy Trade-off

- PLACO Greedy matches Pseudo LB accuracy while querying *only the selected subset* — eliminating cost of unselected humans entirely.
- At budget $B = 0.05 \cdot |H| \cdot K$, PLACO LP maintains high accuracy for larger human pools (where more ideal humans fall within budget).
- Estimation match (fraction of correctly estimated labels) is 25–35%, roughly equal to the fraction of ideal humans — confirming the estimator targets the right cases.
- Posterior-probability estimation outperforms Max-Max, Top-K, and Random baselines on both datasets by a consistent margin.

5.4. Summary Table

Method	Acc. (CNN)	Cost
AI alone (CNN)	56.7%	0
Best human alone	~70%	low
Naïve combination	~82%	all humans
All humans selected	degrades w/ n	all humans
ComHAI (Pseudo LB)	96.7%	all humans
PLACO Greedy	~96.7%	subset only
PLACO LP (budget)	slightly lower	bounded
ComHAI (ResNet-110)	99.2%	all humans

PLACO’s key result: *cost reduction without accuracy loss* — the estimated-label-based subset selection identifies the same high-value humans that true-label-based selection would, at a fraction of the querying cost.